

Data redundancy and inconsistency (Banking)_Test

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

```

79 -- Only account_file is updated
80 -- -----
81
82 • SELECT * FROM account_file;
83

```

account_id	customer_id	customer_name	address	phone	account_type	balance
1	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	Savings	500000.00
2	C002	Su Su	No.22, Hedan Street, Yangon	09-222222222	Current	1200000.00
3	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	Savings	300000.00

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

```

94 • SELECT 'ACCOUNT FILE' AS source, customer_id, customer_name, address
95 FROM account_file
96 WHERE customer_id = 'C001'
97 UNION ALL
98 SELECT 'LOAN FILE' AS source, customer_id, customer_name, address
99 FROM loan_file
100 WHERE customer_id = 'C001'
101 UNION ALL
102 SELECT 'CREDIT CARD FILE' AS source, customer_id, customer_name, address
103 FROM credit_card_file
104 WHERE customer_id = 'C001';

```

source	customer_id	customer_name	address
ACCOUNT FILE	C001	Aung Aung	No.99, Inya Road, Yangon
LOAN FILE	C001	Aung Aung	No.10, Pyay Road, Yangon
CREDIT CARD FILE	C001	Aung Aung	No.10, Pyay Road, Yangon

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

```

211 -- Check consistent result through joins
212 • SELECT
213     c.customer_id,
214     c.customer_name,
215     c.address,
216     a.account_type,
217     a.balance,
218     l.loan_type,
219     l.outstanding_amount,
220     cc.card_number,
221     cc.card_type
222 FROM customers c
223 LEFT JOIN accounts a ON c.customer_id = a.customer_id
224 LEFT JOIN loans l ON c.customer_id = l.customer_id
225 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
226 WHERE c.customer_id = 'C001';

```

customer_id	customer_name	address	account_type	balance	loan_type	outstanding_amount	card_number	card_type
C001	Aung Aung	No.99, Inya Road, Yangon	Savings	500000.00	Home Loan	7500000.00	4111111111111111111	Gold

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

Limit to 5000 rows

```

231 • CREATE VIEW vw_customer_banking_profile AS
232 SELECT
233     c.customer_id,
234     c.customer_name,
235     c.address,
236     c.phone,
237     a.account_id,
238     a.account_type,
239     a.balance,
240     l.loan_id,
241     l.loan_type,
242     l.loan_amount,
243     l.outstanding_amount,
244     cc.card_id,
245     cc.card_number,
246     cc.card_type,
247     cc.credit_limit
248 FROM customers c
249 LEFT JOIN accounts a ON c.customer_id = a.customer_id
250 LEFT JOIN loans l ON c.customer_id = l.customer_id
251 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id;

```

Result Grid

customer_id	customer_name	address	phone	account_id	account_type	balance	loan_id	loan_type	loan_amount	outstanding_amount	card_id
C001	Aung Aung	No.99, Inya Road, Yangon	09-111111111	1	Savings	500000.00	1	Home Loan	10000000.00	7500000.00	1

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

Limit to 5000 rows

```

277 -- Verify inconsistency
278 • SELECT 'ACCOUNT FILE' AS source, address
279 FROM account_file
280 WHERE customer_id = 'C001'
281 UNION ALL
282 SELECT 'LOAN FILE' AS source, address
283 FROM loan_file
284 WHERE customer_id = 'C001'
285 UNION ALL
286 SELECT 'CREDIT CARD FILE' AS source, address
287 FROM credit_card_file
288 WHERE customer_id = 'C001';
289

```

Result Grid

source	address
ACCOUNT FILE	No.99, Inya Road, Yangon
LOAN FILE	No.10, Pyay Road, Yangon
CREDIT CARD FILE	No.10, Pyay Road, Yangon

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

Limit to 5000 rows

```

303 • SELECT
304     c.customer_id,
305     c.customer_name,
306     c.address,
307     a.account_type,
308     l.loan_type,
309     cc.card_type
310 FROM customers c
311 LEFT JOIN accounts a ON c.customer_id = a.customer_id
312 LEFT JOIN loans l ON c.customer_id = l.customer_id
313 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
314 WHERE c.customer_id = 'C001';
315

```

Result Grid

customer_id	customer_name	address	account_type	loan_type	card_type
C001	Aung Aung	No.99, Inya Road, Yangon	Savings	Home Loan	Gold

Difficulty in Accessing Data (Retail Store)_Test

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret... x

Limit to 5000 rows

```
124 -- manual extraction or custom programming is required
125 -- Step 1: View separate files
126 • SELECT * FROM sales_file;
127 • SELECT * FROM stores_file;
128 • SELECT * FROM products_file;
129
130 -- Without a proper integrated design, report logic becomes custom and repetitive
131 • SELECT
132     s.sale_date,
```

Result Grid Filter Rows: Edit Export/Import: Wrap Cell Contents: 15

sale_id	sale_date	store_id	product_id	quantity	total_amount
1	2025-10-05	1	101	10	450000.00
2	2025-10-12	2	102	20	240000.00
3	2025-10-18	3	103	100	80000.00
4	2025-10-22	4	104	60	90000.00
5	2025-10-25	5	105	15	105000.00
6	2025-10-28	6	106	30	75000.00
7	2025-11-03	1	102	25	300000.00
8	2025-11-07	2	101	8	360000.00
9	2025-11-11	3	104	70	105000.00
10	2025-11-15	4	103	120	96000.00

sales_file1 x stores_file2 products_file3 Result 4 Apply Revert

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret... x

Limit to 5000 rows

```
124 -- manual extraction or custom programming is required
125 -- Step 1: View separate files
126 • SELECT * FROM sales_file;
127 • SELECT * FROM stores_file;
128 • SELECT * FROM products_file;
129
130 -- Without a proper integrated design, report logic becomes custom and repetitive
131 • SELECT
132     s.sale_date,
```

Result Grid Filter Rows: Edit Export/Import: Wrap Cell Contents: 15

store_id	store_name	region	city
1	Yangon Central Store	South	Yangon
2	Mandalay Plaza Store	Central	Mandalay
3	Naypyitaw Junction Store	Central	Naypyitaw
4	Mawlamyine Market Store	South	Mawlamyine
5	Taunggyi Hill Store	East	Taunggyi
6	Myitkyina Star Store	North	Myitkyina

sales_file1 stores_file2 x products_file3 Result 4 Apply Revert

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret... x

Limit to 5000 rows

```
124 -- manual extraction or custom programming is required
125 -- Step 1: View separate files
126 • SELECT * FROM sales_file;
127 • SELECT * FROM stores_file;
128 • SELECT * FROM products_file;
129
130 -- Without a proper integrated design, report logic becomes custom and repetitive
131 • SELECT
132     s.sale_date,
```

Result Grid Filter Rows: Edit Export/Import: Wrap Cell Contents: 15

product_id	product_name	category	unit_price
101	Rice Bag	Grocery	45000.00
102	Cooking Oil	Grocery	12000.00
103	Instant Noodles	Grocery	800.00
104	Soft Drink	Beverage	1500.00
105	Shampoo	Personal Care	7000.00
106	Soap	Personal Care	2500.00

sales_file1 stores_file2 products_file3 x Result 4 Apply Revert

Day-1 Assignment

DE_2026_0094

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...

Limit to 5000 rows

```

131 • SELECT
132     s.sale_date,
133     s.store_id,
134     st.store_name,
135     st.region,
136     s.total_amount
137 FROM sales_file s
138 JOIN stores_file st ON s.store_id = st.store_id
139 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
140 ORDER BY st.region, s.sale_date;

```

Result Grid

sale_date	store_id	store_name	region	total_amount
2025-10-12	2	Mandalay Plaza Store	Central	240000.00
2025-10-18	3	Naypyitaw Junction Store	Central	80000.00
2025-11-07	2	Mandalay Plaza Store	Central	360000.00
2025-11-11	3	Naypyitaw Junction Store	Central	105000.00
2025-12-06	2	Mandalay Plaza Store	Central	120000.00
2025-12-10	3	Naypyitaw Junction Store	Central	405000.00
2026-01-08	2	Mandalay Plaza Store	Central	112000.00
2026-01-12	3	Naypyitaw Junction Store	Central	264000.00
2026-02-07	2	Mandalay Plaza Store	Central	125000.00

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

Limit to 5000 rows

```

190 -- Sales report for last 6 months by region
191 • SELECT
192     st.region,
193     SUM(s.total_amount) AS total_sales
194 FROM sales s
195 JOIN stores st ON s.store_id = st.store_id
196 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
197 GROUP BY st.region
198 ORDER BY total_sales DESC;
199

```

Result Grid

region	total_sales
South	2268500.00
Central	2165500.00
East	1229500.00
North	1093500.00

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...)

Limit to 5000 rows

```

201 • SELECT
202     DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month,
203     st.region,
204     SUM(s.total_amount) AS total_sales
205 FROM sales s
206 JOIN stores st ON s.store_id = st.store_id
207 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
208 GROUP BY DATE_FORMAT(s.sale_date, '%Y-%m'), st.region
209 ORDER BY sales_month, st.region;
210

```

Result Grid

sales_month	region	total_sales
2025-10	Central	320000.00
2025-10	East	105000.00
2025-10	North	75000.00
2025-10	South	540000.00
2025-11	Central	465000.00
2025-11	East	100000.00
2025-11	North	84000.00
2025-11	South	396000.00
2025-12	Central	525000.00

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret... x

Limit to 5000 rows

```

212 • SELECT
213     st.region,
214     p.category,
215     SUM(s.total_amount) AS total_sales
216 FROM sales s
217 JOIN stores st ON s.store_id = st.store_id
218 JOIN products p ON s.product_id = p.product_id
219 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
220 GROUP BY st.region, p.category
221 ORDER BY st.region, total_sales DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Content: 11

region	category	total_sales
Central	Grocery	1605000.00
Central	Beverage	337500.00
Central	Personal Care	223000.00
East	Grocery	747000.00
East	Personal Care	482500.00
North	Grocery	777000.00
North	Personal Care	316500.00
South	Grocery	1625000.00
South	Beverage	367500.00

res_file 6 products_file 7 Result 8 Result 9 Result 10 Result 11 x vw_sales_reporting12 sales_file 13 Result 14 Result 15 Result 16 Result 17 Read Only

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret... x

Limit to 5000 rows

```

226 • CREATE VIEW vw_sales_reporting AS
227 SELECT
228     s.sale_id,
229     s.sale_date,
230     st.store_id,
231     st.store_name,
232     st.region,
233     st.city,
234     p.product_id,
235     p.product_name,
236     p.category,
237     s.quantity,
238     s.total_amount
239 FROM sales s
240 JOIN stores st ON s.store_id = st.store_id
241 JOIN products p ON s.product_id = p.product_id;

```

Result Grid Filter Rows: Export: Wrap Cell Content: 11

sale_id	sale_date	store_id	store_name	region	city	product_id	product_name	category	quantity	total_amount
1	2025-10-05	1	Yangon Central Store	South	Yangon	101	Rice Bag	Grocery	10	450000.00
7	2025-11-03	1	Yangon Central Store	South	Yangon	102	Cooking Oil	Grocery	25	300000.00
13	2025-12-02	1	Yangon Central Store	South	Yangon	103	Instant Noodles	Grocery	150	120000.00
19	2026-01-05	1	Yangon Central Store	South	Yangon	104	Soft Drink	Beverage	90	135000.00
25	2026-02-03	1	Yangon Central Store	South	Yangon	105	Shampoo	Personal Care	18	126000.00
31	2026-03-04	1	Yangon Central Store	South	Yangon	106	Soap	Personal Care	60	150000.00
2	2025-10-12	2	Mandalay Plaza Store	Central	Mandalay	102	Cooking Oil	Grocery	20	240000.00
8	2025-11-07	2	Mandalay Plaza Store	Central	Mandalay	101	Rice Bag	Grocery	8	360000.00

res_file 6 products_file 7 Result 8 Result 9 Result 10 Result 11 vw_sales_reporting12 x sales_file 13 Result 14 Result 15 Result 16 Result 17 Read Only

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret... x

Limit to 5000 rows

```

243 • SELECT * FROM vw_sales_reporting;
244
245 -- Test Case Scripts
246 -- Test Case 1: File-based access is difficult
247 -- Raw sales file does not directly show region
248 • USE bad_retaildb;
249 • SELECT
250     sale_id,
251     sale_date,
252     store_id,
253     total_amount
254 FROM sales_file
255 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31';
256
257 -- Test Case 2: Join is required to retrieve region
258

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: 11

sale_id	sale_date	store_id	total_amount
1	2025-10-05	1	450000.00
2	2025-10-12	2	240000.00
3	2025-10-18	3	80000.00
4	2025-10-22	4	90000.00
5	2025-10-25	5	105000.00
6	2025-10-28	6	75000.00
7	2025-11-03	1	300000.00
8	2025-11-07	2	360000.00

res_file 6 products_file 7 Result 8 Result 9 Result 10 Result 11 vw_sales_reporting12 sales_file 13 x Result 14 Result 15 Result 16 Result 17 Apply

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...

Limit to 5000 rows

```

259 • SELECT
260     s.sale_id,
261     s.sale_date,
262     st.region,
263     s.total_amount
264 FROM sales_file s
265 JOIN stores_file st ON s.store_id = st.store_id
266 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31';

```

Result Grid

	sale_id	sale_date	region	total_amount
1		2025-10-05	South	450000.00
2		2025-10-12	Central	240000.00
3		2025-10-18	Central	80000.00
4		2025-10-22	South	90000.00
5		2025-10-25	East	105000.00
6		2025-10-28	North	75000.00
7		2025-11-03	South	300000.00
8		2025-11-07	Central	360000.00
9		2025-11-11	Central	105000.00
10		2025-11-15	South	96000.00
11		2025-11-19	East	100000.00

res_file 6 products_file 7 Result 8 Result 9 Result 10 Result 11 vw_sales_reporting 12 sales_file 13 Result 14 x Result 15 Result 16 Result : Read Only

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...

Limit to 5000 rows

```

269 • USE good_retaildb;
270 • SELECT
271     st.region,
272     SUM(s.total_amount) AS total_sales
273 FROM sales s
274 JOIN stores st ON s.store_id = st.store_id
275 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
276 GROUP BY st.region
277 ORDER BY total_sales DESC;

```

Result Grid

	region	total_sales
1	South	2268500.00
2	Central	2165500.00
3	East	1229500.00
4	North	1093500.00

res_file 6 products_file 7 Result 8 Result 9 Result 10 Result 11 vw_sales_reporting 12 sales_file 13 Result 14 Result 15 x Result 16 Result : Read Only

Query 1 Data redundancy and inconsiste... DDL_DML_DQL_DCL_DTL Difficulty in Accessing Data (Ret...

Limit to 5000 rows

```

281 • SELECT
282     region,
283     COUNT(*) AS transaction_count,
284     SUM(total_amount) AS total_sales
285 FROM vw_sales_reporting
286 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31'
287 GROUP BY region
288 ORDER BY total_sales DESC;
289

```

Result Grid

	region	transaction_count	total_sales
1	South	12	2268500.00
2	Central	12	2165500.00
3	East	6	1229500.00
4	North	6	1093500.00

res_file 6 products_file 7 Result 8 Result 9 Result 10 Result 11 vw_sales_reporting 12 sales_file 13 Result 14 Result 15 Result 16 x Result : Read Only